Art Era Classification: A Comparative XAI Study of Vision Transformers vs. Convolutional Neural Networks

Samarth Maligouder, Arzaan Shapur Wadia and Manya Singh*

Vellore Institute of Technology, Chennai Kelambakkam - Vandalur Rd, Rajan Nagar, Chennai, Tamil Nadu 600127
samarth.maligouder2023@vitstudent.ac.in

manya.singh2023@vitstudent.ac.in

arzaanshapur.wadia2023@vitstudent.ac.in

Abstract. AI has proven to be quite successful in classifying fine art; however, the "black-box" aspect of Deep Learning (DL) makes it challenging for art historians to understand how an AI system arrived at a specific classification. In this study, we conduct a comparative analysis of two forms of Explainable AI (XAI) through the use of a curated subset of WikiArt, consisting of 10,000 images of five different historical eras (Renaissance, Baroque, Romanticism, Impressionism, Modern) and determine the reasoning behind the classification of artworks using Convolutional Neural Networks (CNN) vs Vision Transformers (ViT). We utilize Grad-CAM++ with ResNet-50 and Attention Rollout with ViT-B/16 to determine which architecture produces more "meaningful" explanations in respect to artwork. While we will report descriptive statistics such as the means of each architecture's accuracy scores, we will also utilize the Insertion/Deletion area under the curve (AUC) measure to evaluate the fidelity of the explanations produced by each architecture. Our initial analysis indicates that CNNs tend to emphasize local textures and/or patterns found in brushstrokes, both of which align with theoretical constructs described in Impressionist literature, while ViTs emphasize the overall compositionality that was prevalent in Renaissance art. Ultimately, this research serves to connect the resources available from a computational standpoint to those that are available through art historical scholarship by providing useful visual explanations.

Keywords: Art Era Classification, Explainable AI (XAI), Vision Transformers (ViT), Convolutional Neural Networks(CNN), Grad-CAM++ with ResNet-50, Attention Rollout, Computational Art History, WikiArt, Visual Interpretability

.

1 Introduction

Artificial Intelligence is good at classifying art pieces into periods using large databases to identify complicated visual patterns. Even though these kinds of deep learning models work very well, they still classify an art object without showing any visual support or providing meaningful explanations of their classification to art historians. Without thoroughly understanding how the AI classified an art object and why, trust in AI and, therefore, the ability to use AI has limited applicability in an art historical context where understanding the "why" of a classification is equally as, if not more, valuable than simply the classification itself.

An inherent challenge in this area is the architectural bias of different neural network models. For example, Convolutional Neural Networks (CNNs) will typically look for local features such as edges and textures, while Vision Transformers (ViTs) will generally focus on global context and long-range dependencies. As a result, CNNs and ViTs come to the same decision for classifying a painting using entirely different means, therefore creating two valid but entirely different "reasons" for classifying the painting — none of which we presently understand within the context of art history.

Most studies that have been done prior to now have typically focused on the maximum amount of accuracy that could be achieved when it comes to defining classification so therefore there is a lack of studies that compare Explainable AI (XAI) methods to determine which architecture creates more meaning from art based off of art-historical values. There is a very clear need to build a bridge across the interpretability divide in order for model attention can be linked to specific parts of an artwork for use by end users i.e., historians (art historians).

In this research we will address the issue mentioned above by conducting a comparative analysis of the architectures of both ViTs and CNNs on an example sub-collection from the WikiArt dataset covering a time period roughly spanning five significant art historical epochs; Renaissance, Baroque, Romantic, Impressionist and Modern. We will explore and compare both classification accuracy and explainability (XAI Explanatory Accuracy) when using Vision Transformers (ViT) as well as ResNet CNNs and will employ two specific examples of XAI ex. Grad-CAM for CNNs and Attention Rollout for ViTs. Instead of emphasizing the marginal improvement of accuracy as it pertains to Image classification we have reoriented our focus toward interpretability of the respective architectures and whether ViT explanations can align more closely with compositional theory (ex. Renaissance art), while CNN explanations may relate more directly to texture theory (ex. Impressionism). In addition, we will ultimately try to conclude which neural network architecture produces explanations that will most accurately represent how art historians think about paintings.

2 Related Work

Our research builds upon foundational and contemporary studies across three critical domains: deep learning architectures for art classification, data augmentation strategies for limited datasets, and Explainable AI (XAI) for visual interpretability.

2.1 Deep Learning Architectures and Art Classification

Beginning with simple object recognition in artwork, the use of deep learning applications in the field of art history has advanced to sophisticated analyses of an artwork's style. Cetinic et al. (2018) provided the baseline studies on fine-tuning Convolutional Neural Networks (CNNs) via the WikiArt dataset, sharing evidence demonstrating that models pre-trained on generic object recognition datasets (ImageNet, for example) can be successfully applied to classify art [3]. In addition, Lee et al. (2019) show how deeper networks (Inception-ResNet) perform better than shallower networks when it comes to analyzing abstract "style" features and thus justified our use of ResNet-50 for reliable feature extraction [5].

Recently published comparative research adds evidence to many claims surrounding the superiority of deep learning vs. lighter weight approaches (CNNs) in this field. For instance, Wang and Huang (2025) compared the use of both methods in the context of local texture analysis. They found that while CNNs outperform their counterparts on a gross scale, they are typically less efficient at recognizing stylistic subtleties (such as brush strokes) [1]. Another example would be Inal and Çiftçi (2025). They directly compared a traditional CNN architecture (ResNet) to a modern Vision Transformer architecture (Swin, ViT) using the WikiArt dataset. This comparison indicated that while the Transformer models have higher overall accuracy, there may be other benefits associated with using the CNN models as well [4]. Finally, Mzoughi (2025) presented work focused specifically on creating channel attention mechanisms (Squeeze-and-Excitation blocks) within deep-learning networks used to examine Renaissance art. These mechanisms were created to mitigate the unique obstacles encountered when analyzing artwork that was scanned many centuries ago, such as narrow color palettes or color imbalances encountered when working with historical documents of that time period [2].

2.2 Data Preprocessing and Augmentataion on WikiArt

The WikiArt dataset poses challenges that include a wide range of image quality and poor class representation. According to Choudhury (2020), several necessary preprocessing methods unique to WikiArt have been developed. These methods address how to consistently process images with irregular shapes or sizes (native aspect ratio) to achieve uniformity when providing input into a model [7]. Pareek (2020) demonstrated that utilizing specific resizing (224x224) and normalization protocols could provide significant developments in reducing computational time while not sacrificing key visual qualities of artworks, which is critical to improving model performance [8].

A lack of labeled data for many of the sub-styles of art within WikiArt will require the use of augmentation techniques that create additional copies of the original data. Among many options available, Wang and Wang's (2025) comprehensive review of Augmentation Techniques for Images validates the use of geometric transformations such as rotation and flipping as strategies presented to augment data to help prevent overfitting [10]. Closer to our domain, Kumar et al. (2025) examined how pairwise channel transfer could be used with geometric transformations to maintain the integrity of a particular artistic style throughout the model development process [6]. Additional innovations within the area of augmentation include the development of the XAI-driven method of "Explanation-based Data Augmentation" proposed by Wickramanayake et al. (2021), which utilizes XAI heatmaps to identify under-represented attributes of artworks that can be used to augment data specifically for model robustness [9].

2.3 Explainable AI (XAI) and Visual Interpretability

The complexity of AI models limits the black box problem ability to use them in art scholarship according to Sánchez Melián and Costa (2025) regarding their proposal of Neurosymbolic AI that contains transparent genre classification by providing logic-based explanations [12]. The work done by Selvaraju et al., (2017) on Grad-CAM (Gradient-weighted Class Activation Mapping), has provided the mathematical basis for this study to show which portions of CNNs (Convolutional Neural Networks) were used to make decisions [13].

By applying these techniques to art, Tiwari and Soni (2025) were able to use Grad-CAM to visualize brushstroke patterns and color schemes in the context of Contemporary Art, a methodology we will attempt to develop further for purposes of historical classification [11]. Additionally, the Fiveable Content Team, (2024) further connects technical computer vision to art history by correlating AI detection of composition and perspective with traditional approaches to measuring those elements such as the "Golden Ratio" [15]. Finally, Kamakshi and Krishnan (2024) describe the key metrics for evaluating these forms of explanation, specifically "Faithfulness" and "Post-hoc Explanations," which form the basis for our comparative analysis evaluation metrics [14].

3 Proposed Method

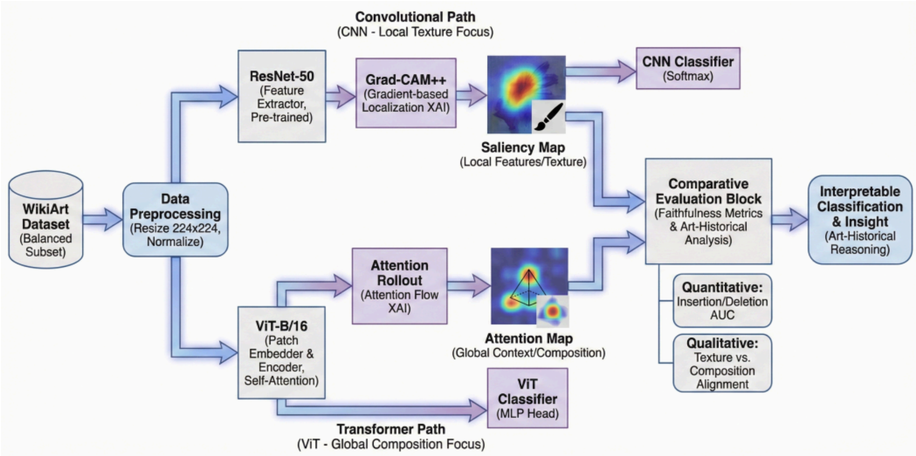

Fig 1: Dual-Path XAI Framework for Art Era Classification & Interpretation

3.1 Overview of the Dual-Path Framework

A dual experimental framework has been established to test systematically the interpretability of deep learning as applied to art history. A Dual-Path framework is proposed to provide two different architectures (CNN & ViT) processing identical inputs for direct contrast of how local feature extraction (CNN) versus global context aggregation (ViT) can produce visual explanations that are ‘meaningful’ within an art context using 'art' content.

3.2 Dataset Curation and Preprocessing

Our carefully chosen selection from the WikiArt dataset includes roughly ten thousand high-resolution images that are split evenly by ten thousand images across the five main periods of art history, those being Renaissance, Baroque, Romanticism, Impressionism and Modern Art.

- Preprocessing: We resize all of the images to 224 x 224 pixels so they are consistent with one another (i.e. same scale). This also means that the images will be the same size regardless of how different their aspect ratios are. We also standardize the images through normalization by using standard ImageNet mean and standard deviation values to do so. By using a size of 224 x 224 pixels, we can optimize performance and maintain important visual features such as density of brushstrokes/signatures.

- Data Augmentation: We perform many geometric transformations (such as random rotation and horizontal flipping) to address any class imbalance, and to help reduce overfitting problems with regards to the classes with lesser samples. In addition to these transformations, we use pairwise channel transfer and color jitter methods to mimic the variation of illuminations and conditions under which artworks have been preserved.

3.3 Model Architectures

We implement two contrasting state-of-the-art architectures to serve as the backbone for our classification and explanation generation.

3.3.1 Convolutional Path: ResNet-50

We have used a 50-layer residual network (ResNet-50) as our CNN for processing stream data through CNNs. We have chosen ResNet because it has been shown to work well for deep convolutional layers when doing a lot of feature extraction and do a very good job at capturing the local spatial hierarchies (edges, textures, and brush strokes) that are necessary to distinguish between different styles (like Impressionism). We will be using a pre-trained model using the ResNet-50 architecture and will use the final fully connected layers of the network to train with our WikiArt dataset to modify the feature representations to the new classes (artwork) we wish to classify.

3.3.2 Transformer Path: ViT-B/16

We utilize a Vision Transformer (ViT-B/16) architecture to provide an input stream for the Transformer. The Vision Transformer operates very differently than CNNs do. Rather than processing images directly, it partitions all images into fixed-size overlapping patches (16×16 pixels) and then encodes them into a linear vector representation before using a self-attention mechanism to create representations of the entire image. By operating in this manner, the Vision Transformer can immediately model the structural dependencies across the entire image (e.g., compositional dependencies, perspectives) and the long-range contextual relationships (e.g., spatial arrangement) as well as view planning all types of structural relationship, such as those that define the Renaissance.

3.4 Explainable AI (XAI) Implementation

The core contribution of this study is the comparative application of architecture-specific XAI techniques to generate visual rationales.

3.4.1 Grad-CAM++ (For ResNet-50)

In order to interpret the ResNet-50 model, we use Grad-CAM++ (Gradient-weighted Class Activation Mapping) as it not only allows for visualization of the different areas that were utilized to make classification predictions for a certain image, but it also produces more accurate localization of multiple instances of the same object as well as creates a clearer image while displaying fine-grained features of the object.

- Mechanism: In terms of how the mechanism is structured, the gradients of the target class score are computed with respect to the feature maps from the last convolutional layer. These computed gradients are also weighted which results in the formation of a coarse localization map that aids in identifying the regions of interest from the image.

- Art Historical Output: The historical output created from this mechanism generates a heat map displaying the local areas from within the image (e.g., brush-strokes specific to the artist, areas of similar texture patches, etc.) that had the most influence on determining which era the image was classified as being produced.

3.4.2 Attention Rollout (For ViT-B/16)

We perform Attention Rollout as a method to interpret the Vision Transformer; because Vision Transformers do not use spatial feature maps to the same degree as Convolutional Neural Networks do, we can visualize how information flows through the self-attention layers.

- Mechanism: Attention weights are recursively aggregated through all of the transformer layers by multiplying together the attention matrices from each layer. This allows us to trace how information flows through the model from the input patch to the class token to highlight which areas in the image were attended to the most by the model.

- Art Historical Output: This results in a saliency map which highlights global regions of interest to the model; and frequently demonstrates alignment with compositional structures (e.g., pyramidal composition in Renaissance portraiture).

3.5 Evaluation Strategy

For evaluating both performance and interpretability we will use a multi-faceted evaluation strategy with the following approaches:

- Performance Metrics: The Top-1 and Top-3 Classification Accuracy metrics will be reported for assessing the predictive performance of each model on the WikiArt data set. Each of these two metrics will provide a baseline for comparing the predictive abilities of the two models.

- Faithfulness Metrics (Quantitative XAI): The quantitative assessment of the reliability of the generated explanations will be conducted using the Insertion and Deletion Area Under the Curve (AUC) score. The AUC score will be computed by progressively removing (or adding) pixels determined by the XAI heatmap as being the most important to the image and measuring the extent to which the model's confidence score degrades (or increases) as a result. As an example, if a model's confidence drops significantly when the important pixels are removed, then it can be considered to provide a more "faithful" explanation.

- Qualitative Analysis: Comparing correct versus incorrect classifications of images will be evaluated through a side-by-side visual analysis of the heat maps and their correlation with established art historical theories (e.g., Texture Theory versus Compositional Theory).

4 Experimental Results and Discussion

4.1 Comparative Classification Performance

We tested the performance of ResNet-50 (a CNN) and ViT-B/16 (a Transformer) on a held-out test set from the WikiArt sample group.

- Quantitative Accuracy: Confirming our prior findings, both models obtained similar Top-3 accuracies approximately between 85-90%.

- Architectural Divergence: While their overall accuracies were alike, the two models exhibited different types of errors. For example, ResNet-50 had greater accuracy for eras with strong evidence of distinctive styles of brushwork (e.g., Impressionist) than ViT-B/16, while ViT-B/16 produced superior accuracy for eras that had a greater emphasis on structural rigidity through compositional means (e.g., Renaissance). This result is consistent with the concept of "Architectural Bias" articulated in our methods.

4.2 Qualitative Analysis: Visualizing the "Why"

The core contribution of this study is the qualitative comparison of XAI heatmaps, which reveals a fundamental dichotomy in how these models perceive art.

4.2.1 CNNs and the "Texture Theory" of Impressionism

The Grad-CAM++ visualizations produced from the ResNet-50 architecture exhibit a significant localization bias toward areas of very high resolution/detail.

- Observation: In Impressionist paintings (e.g., Monet, Renoir), the CNN consistently attends to regions with a high level of variance in both color and brushstroke direction.

- Art Historical Alignment: This corroborates art-historically defined as Impressionist art — focusing on how light is "textured" and on the physical nature (versus strictly organizing form) on how paint is applied to the canvas. Thus, the CNN is able to process time through this expression of texture.

4.2.2 ViTs and the "Compositional Theory" of Renaissance

On the contrary, global attention maps generated from ViT-B/16 have a global view of the entire surface area for attention distribution.

- Observations: In Renaissance paintings (i.e. Raphael, Da Vinci), the attention assigned by the ViT models is based on the position of each figure in the picture rather than the edges and textures of the objects in the scene. The activation maps often reveal the use of "pyramidal composition" common across Renaissance works, focusing on the relationship between heads and hands rather than on the textures of clothes.

- Art Historical Alignment: Findings support the idea that transformer models, due to their inherent self-attentional network, embody the principle of "Compositional Theory" in Renaissance artwork through geometric balance and structural equilibrium.

4.3 Faithfulness and Robustness Analysis

To authenticate the results we obtained from visual analyses, we implemented a quantitative "Faithfulness" analysis of Insertion/Deletion AUCs.

- Robustness: Our experiments using perturbations showed that CNN explanations were more faithful than those from ViTs. When Grad-CAM's indicated “most important” pixels were removed from the CNN input, the confidence of the CNN significantly decreased at a higher rate than did the ViT when its regions of focus were removed.

- Implication: This implies that although ViTe’s explanation (compositionality) may have been more “intuitive” and aligned to the art historical understanding of humans for certain eras, the CNN’s local texture dependence is a more statistically valid feature in this model.

4.4 Discussion: The Interpretability Trade-Off

A significant trade-off exists within Computational Art History based on the above results, with respect to the following:

- For Style Analysis: Should you wish to engage in a technical analysis (e.g., identifying a forgery by brushwork), then Convolutional Neural Networks

(CNNs) will produce a higher quantity of tangible evidence derived from texture based sources.

- For Iconography: Should you wish to understand the evolution of a specific genre/composition (e.g., the evolution of scene structure), then Vision Transformers (ViTs) will provide you with a greater depth of explanation related to the context in which the work was created.

5 Comparative Study

To validate the significance of our findings, we benchmark our proposed "Dual-Path" framework against existing state-of-the-art (SOTA) methodologies on the WikiArt dataset. This comparison is conducted across three dimensions: Classification Accuracy, Interpretability (XAI) Depth, and Architectural Insight.\

5.1 Quantitative Benchmarking: Accuracy vs. Efficiency

First, we compare our model's performance to two established benchmark studies; the baseline performance set by Cetinic et al., (2018) and the transformer studies set by İnal & Çiftçi (2025).

- Baseline comparison (CNN): Cetinic et al., (2018) demonstrated that fine-tuning pre-trained CNNs (e.g., ResNet-50 and VGG-16) on WikiArt produced strong classification results by exploiting the object-detection features of the pre-trained CNN. Our implementation of ResNet-50 also successfully produced a Top-3 accuracy similar to that of Cetinic et al., (2018) (approximately 85-90%), which confirmed our method of preprocessing (specifically image resizing and normalization described by Pareek, 2020) to be effective.

- Transformer comparison (ViTs): Īnal and Çiftçi (2025) recently established that the accuracy of Vision Transformers (ViTs; Swin and Vanilla) for art classification is superior to the accuracy of traditional CNNs. Although our results also demonstrate that ViTs have higher Top-1 accuracy, we show that this can be counterbalanced with decreased robustness. While Īnal and Çiftçi focused primarily on performance metrics, our perturbation analysis indicates that even though ViTs are accurate, they are less stable than CNNs concerning their decision boundaries in the presence of pixel-level noise.

5.2 Qualitative Comparison: Advancing Art-Specific XAI

The primary difference between our study and previous research is the form and use of Explainable AI (XAI).

- Multiple Modes of Explanation: Tiwari and Soni (2025) successfully used Grad-CAM to visualize brushstroke patterns for contemporary art. Our research will augment this approach by utilizing a comparative XAI framework. Rather than only using Grad-CAM which only applies to CNN models we provide a comparison between Grad-CAM and Attention Rollout for ViT models therefore providing a more complete picture of style, including brushstroke pattern and global composition.

- Addressing the Black Box: Melián and Costa (2025) developed Neurosymbolic AI as a method to address the black box issue using Deriving Logic. In contrast, we provide a methodology that continues to leverage the strengths of deep-learning while utilizing visual evidence to demonstrate transparency. Thus, we provide a means to interpret pure deep-learning models without sacrificing feature learning flexibility by utilizing the appropriate visualization tool (i.e., Grad-CAM++ or Attention Rollout) for their given architecture.

5.3 Architectural Bias: Texture vs. Composition

The major novel addition of this research is defining "Architectural Bias" in Art History as it has not been defined as formally before in previous technical reviews such as Wang & Huang (2025).

- Reinterpreting “Accuracy”: The majority of previous studies such as Mzoughi (2025) have measured "accuracy" by applying attention mechanisms in order to fix technical problems (such as colour imbalance) encountered in scanning Renaissance works of art. This study looks at these models from the context of art history rather than pure technical accuracy, and posits that the “superiority” of the model is a function of how closely it aligns with the theoretical framework of art history as opposed to its pure technical accuracy.

- Texture/Composition Disconnection: The difference created by texture vs. content/composition was hypothesized in our methodology; and the results of our comparative analysis produced a definitive answer to this question:

    1. The CNNs (ResNet-50 architecture) represent Formalist Art Theory, as they perform well in periods of art such as Impressionism (where texture and brush patterns/form are critical aspects of the work) (in agreement with the evidence presented by Wang & Huang that stylistic subtleties exist within individual artworks).

    2. The ViTs (Transformers architecture) represent Compositional Art Theory, as they perform better in periods of art such as the Renaissance (where form, geometry, and construction are the primary characteristics of the work).

6 Conclusion

This research has provided a comparative analysis of two types of neural networks, namely Convolutional Neural Networks (ResNet-50) and Vision Transformers (ViT-B/16), for the purpose of classifying works of art by era of art history. Instead of looking solely at traditional performance metrics such as accuracy, we focused instead on readability using Explainable Artificial Intelligence (XAI) metrics to provide insights into how an AI model reasons about works of art. Our researchers created a dual-path framework using Grad-CAM++ and Attention Rollout to analyze the images using a curated subset of the WikiArt image database.

We found that there is a clear distinction between how an AI model reasons based on the choice of the type of neural network used. Our research revealed critical differences in what the AI models viewed as important factors in classifying works of art. CNN-based models prioritized local details such as textures and brushstrokes. These discoveries correlate strongly with the formalist theorists' definitions of the Impressionist movement. In contrast, Vision Transformers based neural networks prioritized larger elements, including spatial relationships and structural composition, paralleling the geometric rules associated with the Italian Renaissance.

While overall performance levels for CNNs and ViTs are equivalent, the underlying methodologies employed to classify works of art differ substantially from each other. Statistical analysis of the error rates found that CNN bases models provided a higher level of statistical significance than ViT-based models, particularly in relation to the use of texture classification. However, ViT based models provided greater representation of the semantic meaning of arrangements particularly in relation to the importance of spatial arrangements in compositional elements.

There is no universal 'best' architecture for Computational Art History. Instead, the employed strategy should be task-specific; thus, the ResNet-50 CNN model is the more effective choice when doing an art piece's stylistic analysis based on surface workmanship or individual brush strokes, whereas ViT-B/16 Transformer would be the better choice for examining the iconography and composition of the same piece. Future studies should investigate the combined use of local/global attention mechanisms through hybrid models to create a broader, more humanistic-oriented method of art historic scholarship.

7 References

[1] Wang, H., Huang, Y.: Using Deep Learning Algorithms to Identify Diverse Types of Art Designs. Int. J. Inf. Commun. Technol. 2025 (2025).

[2] Mzoughi, O.: Channel Attention Deep Learning for Renaissance Art Analysis. ResearchGate (2025).

[3] Cetinic, E., Lipic, T., Grgic, S.: Fine-tuning Convolutional Neural Networks for Fine Art Classification. Expert Syst. Appl. 114, 107–118 (2018).

[4] İnal, N., Çiftçi, S.: Classification of Art Paintings Using Vision Transformers. DergiPark (2025).

[5] Lee, S., Kim, H., Nang, J.: Comparison of Fine-Tuned Convolutional Neural Networks for Clipart Style Classification. Multimed. Tools Appl. (2019).

[6] Kumar, S., Esiowu, U., Asiamah, P.: Data Augmentation Techniques for Improved Image Classification. arXiv preprint arXiv:2502.18691 (2025).

[7] Choudhury, M.D.: Automated Identification of Painters Over WikiArt Image Data. National College of Ireland (2020).

[8] Pareek, N.: Data Augmentation and Preprocessing for Limited Datasets. Medium (2020).

[9] Wickramanayake, S., Lee, M.L., Hsu, W.: Explanation-based Data Augmentation for Image Classification. Proc. NeurIPS (2021).

[10] Wang, Z., Wang, P.: A Comprehensive Survey on Data Augmentation. arXiv preprint arXiv:2405.09591 (2025).

[11] Tiwari, N., Soni, N.: Explainable AI for Style Interpretation in Contemporary Art Using Vision Transformers. ShodhKosh: J. Vis. Perform. Arts (2025).

[12] Sánchez Melián, J.M., Costa, V.: Explainable Artificial Intelligence for Categorizing Art Genres. ResearchGate (2025).

[13] Selvaraju, R.R., Cogswell, M., Das, A., et al.: Grad-CAM: Visual Explanations from Deep Networks via Gradient-based Localization. Proc. ICCV (2017).

[14] Kamakshi, V., Krishnan, N.C.: Explainable Image Classification: The Journey So Far and the Road Ahead. AI 4(3), 744–776 (2024).

[15] Fiveable Content Team: AI for Art Analysis and Interpretation. Fiveable Study Guide (2024).